TEEM-BH 2.0

Energetic Black Hole Physics: Horizons, Cores, Entropy, and Information

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Abstract

Black holes provide the most extreme environment in which to test any fundamental theory of physics. Within the Triple Energy Exchange Model (TEEM3.0), spacetime is described not by geometric curvature but by the energetic configuration of the spatial tension field R. Time flow is governed by the ratio v/R, and gravity emerges from gradients of the combined energetic potential $m + R$. Applying this framework to strong gravity, we develop TEEM-BH, an energetic description of black hole horizons, interiors, entropy, and information flow.

We show that the event horizon corresponds to an **R-critical surface** where $v/R \to 0$, producing an energetic freeze-out that reproduces the causal structure of General Relativity while providing a physical mechanism for horizon formation. Gravitational collapse drives the tension field toward a universal saturation value, forming a **finite-tension core** that replaces the classical singularity. Black hole entropy arises from **boundary tension modes** on the R-critical surface, yielding the area law and a physical interpretation of the Bekenstein–Hawking coefficient as the density of tension modes. Information is preserved through **R-reconfiguration**: infalling matter transfers its information into the tension field, which stores it in long-memory configurations and gradually releases it through correlated, non-thermal Hawking-like radiation.

TEEM-BH predicts several observational signatures distinct from General Relativity, including a finite horizon thickness, modified shadow profiles, gravitational-wave echoes from the R-critical layer, and near-horizon anisotropies arising from R-memory. These effects are testable with current and upcoming facilities such as the Event Horizon Telescope, LIGO/Virgo/KAGRA, and LISA. By grounding black hole physics in energetic rather than geometric principles, TEEM-BH provides a unified, non-singular, and observationally falsifiable framework for strong gravity.

1. Introduction

Black holes represent the most extreme physical environments known in the Universe. Near their horizons, **time flow slows**, **gravity becomes dominant**, and **quantum information must remain unitary**. Any fundamental theory of physics must therefore explain black hole horizons, interior structure, entropy, and information flow in a way that is both physically meaningful and observationally testable.

General Relativity (GR) provides an elegant geometric description of black holes, predicting the existence of event horizons and describing the external gravitational field with remarkable accuracy. However, GR also predicts **geometric singularities**, regions where curvature diverges and the theory ceases to be valid.

Furthermore, GR offers no microscopic explanation for **black hole entropy**, nor a mechanism for **information preservation** during evaporation. These limitations indicate that the geometric description of spacetime, while powerful, may be incomplete in the strong-gravity regime.

The Triple Energy Exchange Model (TEEM3.0) offers an alternative foundation. In TEEM, spacetime is not geometric but **energetic**, characterized by the spatial tension field R. Energy exists in three interchangeable modes—vibrational v, condensed m, and spatial tension R— and physical phenomena arise from their redistribution. Time flow is governed by the ratio v/R, and gravity emerges from gradients of the combined energetic potential $m + R$. This framework reproduces the tested predictions of GR while providing a deeper physical interpretation of gravitational behavior, temporal flow, and horizon formation.

Black holes are the natural testing ground for TEEM. Near a horizon, the ratio v/Rapproaches zero, causing time to freeze; the tension field Rrises sharply and saturates; and mode conversion $v, m \to R$becomes dominant. These features suggest that black hole horizons and interiors can be understood as **energetic phase transitions** rather than geometric singularities.

The purpose of this paper is to develop **TEEM-BH**, the black hole branch of the TEEM3.0 framework. We show that:

- the event horizon corresponds to an **R-critical surface** where $v/R \to 0$;
- gravitational collapse leads to a **finite-tension core**, avoiding singularities;
- black hole entropy arises from **boundary tension modes** on the horizon;
- information is preserved through **reconfiguration of the R-field**;
- and TEEM predicts **observable deviations** from GR in shadow profiles, gravitational-wave echoes, and Hawking-like radiation.

By grounding black hole physics in energetic rather than geometric principles, TEEM-BH provides a unified, non-singular, and observationally testable description of strong gravity. It extends TEEM cosmology into the black hole regime and offers a coherent energetic interpretation of horizons, entropy, and information flow.

2. Minimal TEEM Foundations (Self-Contained Summary)

The Triple Energy Exchange Model (TEEM3.0) describes all physical phenomena through the redistribution of a single conserved quantity—energy—across three interchangeable modes: vibrational energy v, condensed energy m, and spatial tension energy R. Spacetime is not geometric but **energetic**, with Rrepresenting the intrinsic tension of space. This section summarizes the minimal TEEM principles required for the black hole analysis that follows.

2.1 Three Fundamental Energy Modes

TEEM decomposes total energy into a linear combination:

$$E = \alpha v + \beta m + \gamma R.$$

Each mode has a distinct physical meaning:

- **Vibrational energy** v Oscillatory degrees of freedom: radiation, quantum fluctuations, kinetic excitations. Governs temporal evolution.
- **Condensed energy** m Localized, mass-like energy: matter, rest mass, coarse-grained internal energy.
- **Spatial tension energy** R The energetic state of space itself. Controls gravitational behavior, refractive properties, and the rigidity of spacetime. Evolves slowly and retains long-range memory.

Energy continuously converts among the modes:

$$v \leftrightarrow m \leftrightarrow R,$$

with total energy conserved.

2.2 Time Flow as an Energetic Ratio

In TEEM, time is not a geometric coordinate but an emergent property of the energetic state:

$$\frac{d\tau}{dt} = \frac{v}{R}.$$

- High v/ low R→ **fast time flow**
- Low v/ high R→ **slow time flow**
- $v/R \to 0$→ **time freeze-out**

This definition reproduces gravitational and relativistic time dilation without invoking curvature. Horizons naturally arise as surfaces where v/R becomes vanishingly small.

2.3 Gravity as an Energetic Gradient

Gravitational acceleration emerges from spatial variations in the combined energetic potential:

$$g = -\nabla(m + R).$$

- m contributes as in classical gravity
- R contributes as a distributed tension field
- No geometric curvature is required

This formulation reproduces Newtonian gravity and GR predictions in tested regimes, while providing a physical origin for dark-matter-like effects through gradients in R.

2.4 Energetic Spacetime and Partial-Reset Dynamics

Spatial tension R evolves slowly and retains memory of past energetic configurations. However, TEEM3.0 does **not** require a "previous universe" or external initial conditions. Instead, extreme energetic states trigger a **partial-reset**:

- R saturates at a finite maximum
- mode conversion redistributes energy
- singularities are avoided
- information is preserved in the configuration of R

This mechanism applies both to cosmological bounces and to black hole interiors.

2.5 Why TEEM is Suited for Strong Gravity

Black holes naturally push the TEEM energy modes to their limits:

- R rises sharply near collapse
- v/R approaches zero at the horizon
- mode conversion $v, m \to R$ becomes dominant
- R-saturation prevents singularities
- R-memory stores information

3. Horizon as an R-Critical Surface

In General Relativity, the event horizon is defined geometrically as a null surface beyond which no causal signal can escape. In TEEM, this geometric definition is replaced by an **energetic** one. A black hole horizon forms where the spatial tension field R reaches a critical value that suppresses all vibrational activity, causing the local flow of time to freeze.

3.1 Definition of the R-Critical Horizon

TEEM defines the horizon as the surface where the spatial tension energy reaches its universal saturation value:

$$R(r_h) = R_{\text{crit}}.$$

At this surface, the ratio governing time flow,

$$\frac{d\tau}{dt} = \frac{v}{R},$$

approaches zero:

$$\frac{v}{R_{\text{crit}}} \;\longrightarrow\; 0.$$

Thus, **the TEEM horizon is the surface where time effectively halts**. This definition is not geometric but **energetic**, arising directly from the fundamental TEEM relation between vibrational energy and spatial tension.

3.2 Time Freezing and Energetic Causal Disconnection

Because information is carried by vibrational energy v, the condition $v/R \to 0$ has immediate physical consequences:

- vibrational modes cannot propagate outward,
- local temporal evolution becomes arbitrarily slow,
- the region becomes causally isolated.

In TEEM, causal disconnection is therefore an **energetic freeze-out**, not a geometric trapping of null geodesics. The horizon marks a **phase boundary** in the energetic configuration of space.

3.3 Behavior of the Tension Field Near the Horizon

As matter collapses, condensed energy m increases the local tension field R through mode conversion. The gradient of the combined potential $m + R$ produces gravitational acceleration:

$$g = -\nabla(m + R).$$

Near the horizon, this gradient becomes extremely steep, driving R rapidly toward its critical value. The tension field exhibits a characteristic profile:

- smooth at large radii,
- sharply rising near r_h,
- saturating at R_{crit} inside the horizon.

This behavior mirrors a **second-order phase transition** in the energetic structure of space.

3.4 Correspondence with the Schwarzschild Radius

Although TEEM replaces geometric curvature with energetic tension, the location of the R-critical surface coincides with the classical Schwarzschild radius:

$$r_h \approx r_s.$$

This correspondence arises because the condition $v/R \to 0$ occurs precisely where GR predicts infinite redshift. Thus TEEM preserves the successful predictions of GR while providing a **physical mechanism**—tension saturation—for horizon formation.

3.5 Horizon as an Energetic Phase Boundary

The TEEM horizon is not a geometric singularity but a **critical surface** characterized by:

- **R-saturation**: $R = R_{\mathrm{crit}}$
- **time freeze-out**: $v/R \to 0$
- **suppressed mode conversion**: $v, m \nrightarrow R$ beyond saturation
- **boundary tension modes**: microscopic degrees of freedom living on the surface
- **energetic causal isolation**: no outward propagation of vibrational energy

This interpretation unifies horizon physics with TEEM cosmology, where similar saturation phenomena prevent singularities during cosmic contraction.

3.6 Figure 1: R-Critical Horizon Profile

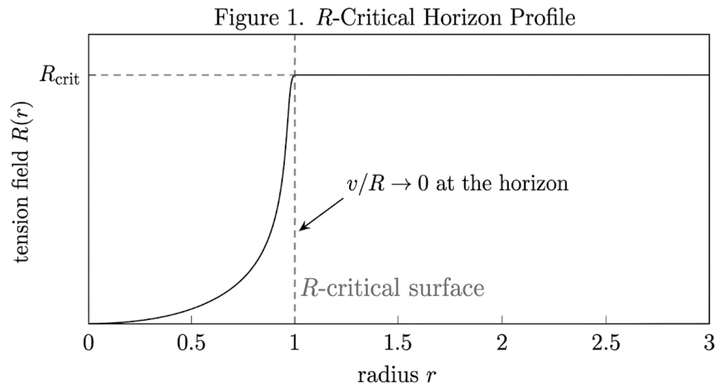

Figure 1. R-Critical Horizon Profile

Radial dependence of the spatial tension field $\boldsymbol{R(r)}$ near the horizon. The tension field rises steeply and saturates at $\boldsymbol{R_{\text{crit}}}$, defining the R-critical surface where $\boldsymbol{v/R \to 0}$ and time flow freezes.

4. Finite-Tension Core and Singularity Avoidance

General Relativity predicts that gravitational collapse inevitably leads to a singularity—a region where curvature diverges and physical quantities become ill-defined. In TEEM, this outcome is avoided because the spatial tension field R cannot grow without bound. Instead, collapse drives R toward a universal saturation value R_{crit}, producing a **finite-tension core** rather than a geometric singularity.

4.1 Collapse Dynamics in TEEM

During gravitational collapse, the three TEEM energy modes evolve in a characteristic sequence:

1. **Condensed energy** m increases as matter density rises.
2. **Vibrational energy** v is suppressed due to the increasing tension field.
3. **Spatial tension** R grows rapidly through mode conversion:

$$m \longrightarrow R.$$

The gravitational field is determined by the gradient of the combined energetic potential:

$$g = -\nabla(m + R),$$

so the steep rise of R near the center enhances the inward acceleration, further accelerating collapse. However, unlike curvature in GR, R **has a physical upper bound**.

4.2 R-Saturation Mechanism

TEEM imposes a universal limit on the spatial tension field:

$$R \leq R_{\text{crit}}.$$

When R approaches this critical value:

- further conversion $m \to R$ becomes energetically suppressed,
- the growth of R slows dramatically,

- excess condensed energy is redirected into internal tension modes or residual vibrational modes.

This saturation condition can be expressed as:

$$\frac{dR}{dt} \longrightarrow 0 \text{ as } R \to R_{\text{crit}}.$$

Thus, **collapse cannot drive R to infinity**, and no physical quantity diverges.

4.3 Formation of a Finite-Tension Core

Once R reaches R_{crit}, the interior enters a stable, high-tension state. The result is a **finite-sized core** characterized by:

- $R \approx R_{\text{crit}}$ throughout the interior,
- **finite density** (no divergence),
- **extremely slow but nonzero time flow** ($v/R \ll 1$),
- **suppressed but nonvanishing vibrational modes**,
- **internal tension modes storing information**.

This core is not a geometric point but an extended region whose radius depends on the black hole mass and the conversion efficiencies α, β, γ.

It is the **energetic endpoint of collapse**, replacing the singularity of GR.

4.4 Comparison with GR Singularities

Feature	GR	TEEM-BH
Curvature	Diverges	Finite (R saturates)
Density	Infinite	Finite
Time flow	Undefined	Extremely slow but defined
Interior structure	Singular point	Finite-tension core
Information	Lost or ambiguous	Stored in R-configuration

TEEM therefore resolves the singularity problem by replacing geometric divergence with **energetic saturation**.

4.5 Relation to TEEM Cosmology (No "Previous Universe")

TEEM3.0 eliminates the need for a "previous universe" or external initial conditions. Instead, both cosmological bounces and black hole interiors follow the same **partial-reset mechanism**:

- R rises toward a maximum,
- saturation halts divergence,
- energy redistributes among modes,
- information is preserved in the configuration of R.

Thus, the black hole interior is a finite R-saturation region analogous to a small cosmological bounce, but it does not require any remnant of a previous universe.

4.6 Figure 2: Finite-Tension Core Structure

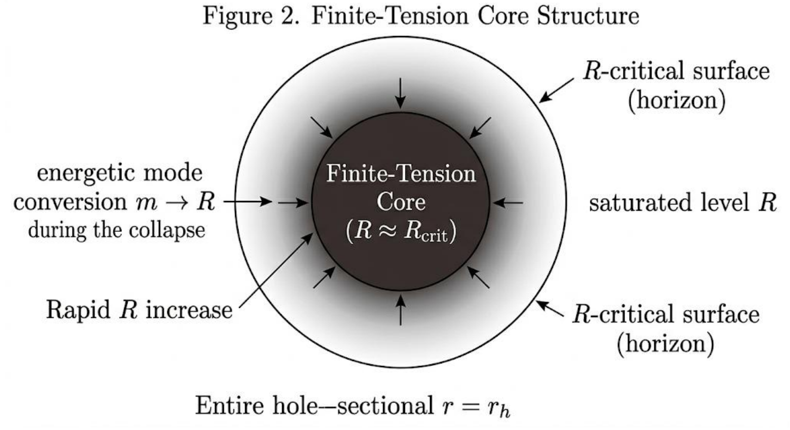

Figure 2. Finite-Tension Core Structure

Schematic cross-section of a collapsed object showing the finite-tension core ($\boldsymbol{R} \approx \boldsymbol{R}_{\text{crit}}$), the rapid increase of $\boldsymbol{R}$ near the horizon, and energetic mode conversion $\mathbf{m} \to \boldsymbol{R}$ during collapse. The R-critical surface defines the horizon where the tension field saturates.

5. Entropy from Boundary Tension Modes

Black hole entropy provides a profound clue about the microscopic structure of spacetime. In General Relativity, the Bekenstein–Hawking formula establishes that entropy is proportional to the area of the event horizon, yet GR offers no physical explanation for the origin of these degrees of freedom. In TEEM, the answer arises naturally: **entropy is the count of microscopic tension modes living on the R-critical surface**.

5.1 Horizon as a Tension-Dominated Membrane

At the R-critical surface, vibrational energy is suppressed:

$$\frac{v}{R_{\text{crit}}} \to 0,$$

so the only remaining dynamical degrees of freedom are **fluctuations of the spatial tension field** R. The horizon therefore behaves as a **tension-dominated membrane**, characterized by:

- high spatial tension,
- suppressed vibrational activity,
- localized boundary excitations,
- large degeneracy of microstates.

This is not an analogy—the membrane is a literal energetic surface defined by the saturation of R.

5.2 Boundary Tension Modes as Microstates

Because R saturates at the horizon, it cannot increase further, but **small tangential fluctuations of** R along the surface remain energetically allowed. These fluctuations form discrete, quasi-independent **boundary tension modes**.

Let each minimal patch of area ΔAsupport one independent mode. Then the number of microstates is:

$$\Omega = 2^{A/\Delta A}.$$

The entropy is therefore:

$$S = k_B \ln\, \Omega = \frac{k_B}{\Delta A} A.$$

Thus, the **area law emerges directly** from counting the number of tension modes on the horizon.

5.3 Area Law and the Bekenstein–Hawking Coefficient

Matching TEEM's expression for entropy with the Bekenstein–Hawking formula,

$$S_{\mathrm{BH}} = \frac{k_B c^3}{4G\hbar} A,$$

implies that the fundamental area scale ΔAcorresponds to the **density of R-modes** on the horizon:

$$\Delta A = \frac{4G\hbar}{c^3}.$$

In TEEM, this is not a geometric Planck area but the **minimum resolvable patch of spatial tension**. The entropy coefficient therefore acquires a clear physical meaning: it reflects the **quantization of the tension field**, not the quantization of geometry.

5.4 Physical Interpretation

This energetic interpretation resolves several conceptual issues:

- **Why entropy scales with area, not volume:** because only boundary tension modes remain active at $R = R_{\mathrm{crit}}$.
- **Why the horizon behaves like a thermodynamic system:** because the tension field supports a large number of microstates.
- **Why holography emerges naturally:** information is stored in the configuration of R on the boundary.
- **Why entropy is universal across black hole types:** the R-critical condition is universal.

TEEM therefore provides a microscopic, physically motivated origin for black hole entropy without invoking strings, loops, or geometric quantization.

5.5 Figure 3: Boundary Tension Modes

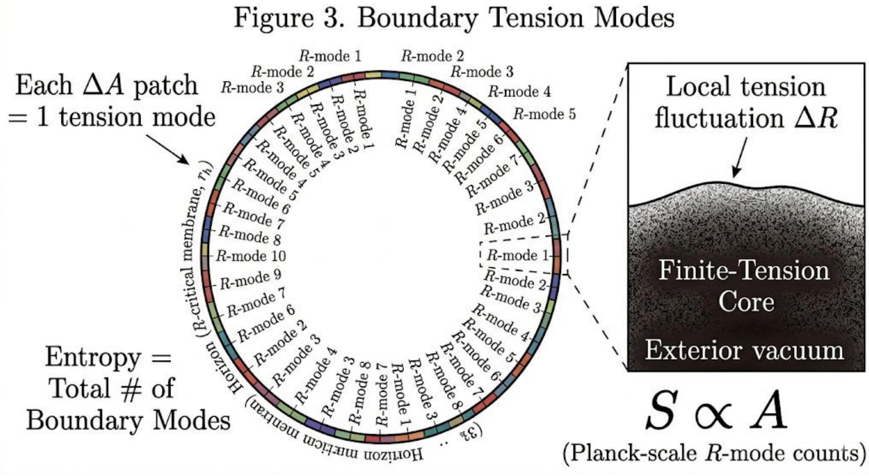

Figure 3. Boundary Tension Modes

Schematic representation of microscopic tension modes distributed along the R-critical surface. Each patch of area $\boldsymbol{\Delta A}$ corresponds to one independent R-mode, and the total number of boundary modes determines the entropy $\boldsymbol{S \propto A}$. Local tension fluctuations $\boldsymbol{\Delta R}$ on the horizon represent Planck-scale degrees of freedom.

6. Information Preservation via R-Reconfiguration

The black hole information paradox arises from the apparent conflict between quantum unitarity and the classical prediction that information falling into a black hole is irretrievably lost. In TEEM, this conflict does not arise. Because the spatial tension field R is an energetic degree of freedom with internal structure and long-memory, information is not destroyed but **reconfigured** into the pattern of the tension field at and within the horizon.

6.1 Mode Conversion as Information Transfer

As matter or radiation approaches the horizon, the rising tension field suppresses vibrational activity:

$$\frac{v}{R} \to 0.$$

This suppression triggers a directional mode conversion:

$$(v, m) \;\longrightarrow\; R.$$

Crucially, this conversion is **not erasure**. The TEEM3.0 conservation principle ensures that the total information stored across the three modes remains constant:

$$J_v + J_m + J_R = \text{constant}.$$

Thus, information originally encoded in vibrational or condensed modes is transferred into the **configuration of the tension field**.

6.2 R-Configuration as an Information-Bearing Field

The spatial tension field $R$is not a scalar quantity but a structured energetic field with:

- local tension gradients,
- anisotropies induced by infalling matter,
- boundary tension mode patterns,
- residual vibrational imprints suppressed but not erased,
- slow evolution and long-memory.

These features allow $R$to encode a vast amount of information. The horizon acts as an **information boundary**, where incoming data is reorganized into boundary tension modes, while the interior finite-tension core stores deeper structural information.

In TEEM, holography emerges naturally: **information resides on the boundary because the boundary is where R-modes remain active.**

6.3 Hawking-Like Radiation as R-Relaxation

In TEEM, Hawking radiation is not produced by pair creation at the horizon. Instead, it arises from the **slow relaxation of the tension field**:

$$R \longrightarrow v.$$

As the black hole evaporates:

- the tension field gradually decreases,
- stored information is released as vibrational energy,
- outgoing radiation carries **correlations** reflecting the original R-pattern.

This radiation is therefore **not perfectly thermal**. It contains subtle structure encoding the history of infalling matter.

TEEM thus provides a physically transparent, unitary mechanism for black hole evaporation.

6.4 No-Loss Principle and Unitarity

TEEM enforces a strict **No-Loss Principle**:

- information may change form,
- but it is never annihilated.

Because the tension field Rsaturates rather than diverges, and because its configuration retains long-memory, the total information content of the system remains conserved throughout collapse, horizon formation, and evaporation.

This resolves the information paradox without requiring:

- firewalls,
- nonlocal wormholes,
- exotic quantum gravity effects,
- or violations of semiclassical physics.

6.5 Unified Picture of Information Flow

The TEEM information cycle can be summarized as:

1. **Infall:** $(v, m) \to R$— information transferred into tension field.
2. **Storage:** R-configuration patterns persist in the finite-tension core and boundary modes.
3. **Evaporation:** $R \to v$— information released as correlated Hawking-like radiation.
4. **Conservation:** $J_v + J_m + J_R$ = constant.

Thus, black hole evolution is fully unitary.

6.6 Figure 4: Information Flow Diagram

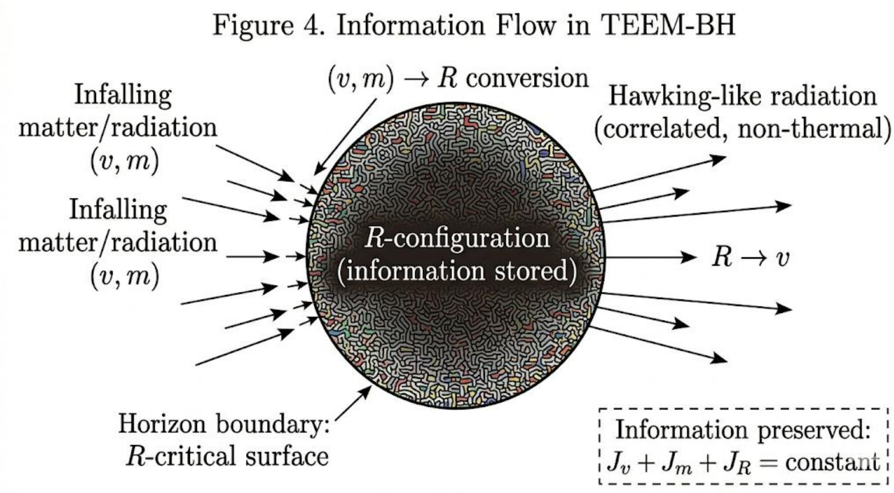

Figure 4. Information Flow in TEEM-BH

Schematic representation of information transfer and preservation in TEEM-BH. Infalling matter and radiation $(\boldsymbol{v}, \boldsymbol{m})$ *are converted into the tension field* $\boldsymbol{R}$ *at the horizon boundary (R-critical surface). The R-configuration stores information internally and gradually releases it through correlated, non-thermal Hawking-like radiation as* $\boldsymbol{R} \to \boldsymbol{v}$*. Total information is conserved:* $\boldsymbol{J_v} + \boldsymbol{J_m} + \boldsymbol{J_R}$ = constant.

7. Observable Predictions

A physical theory of black holes must produce testable predictions that distinguish it from General Relativity (GR) while remaining consistent with existing observations. TEEM-BH yields several such predictions, arising from its energetic interpretation of horizons, the saturation of the tension field R, and the slow relaxation of R-configurations. These signatures span black hole imaging, gravitational waves, and Hawking-like radiation.

7.1 Finite Horizon Thickness

In TEEM, the horizon is not an infinitesimally thin null surface but a **finite-width transition region** where the tension field rises sharply toward its critical value:

$$\Delta r_h \sim \left(\frac{dR}{dr}\right)_{r_h}^{-1}.$$

Consequences:

- the photon ring should exhibit a **smoother brightness gradient** than predicted by GR,
- the shadow boundary becomes **slightly blurred**,
- the effective refractive index near the horizon is modified by the steep R-gradient.

EHT and next-generation VLBI should be able to detect these deviations.

7.2 Modified Black Hole Shadow Profile

Because the horizon is an R-critical surface rather than a geometric null surface, the refractive properties of space near the horizon differ from GR.

TEEM predicts:

- a **slightly larger shadow diameter** for a given mass,
- a **less sharply defined photon ring**,
- potential **azimuthal asymmetries** if the R-field retains memory of anisotropic infall.

These effects arise from the energetic structure of R and its slow-memory, and they fall within the sensitivity of upcoming EHT observations of Sgr A* and M87*.

7.3 Non-Thermal Components in Hawking-Like Radiation

In TEEM, Hawking radiation originates from **R-relaxation**, not pair creation. As the tension field slowly decreases:

$$R \to v,$$

the emitted radiation carries **correlations** reflecting the stored R-configuration.

Predictions:

- deviations from a perfect blackbody spectrum,
- long-range correlations between emission events,
- a slower evaporation rate for small black holes.

Future gamma-ray telescopes and primordial black hole searches could detect these signatures.

7.4 Gravitational-Wave Echoes from the R-Critical Layer

The R-critical surface acts as a partially reflective energetic boundary. Perturbations generated during black hole mergers can scatter off this layer, producing **late-time echoes** in the gravitational-wave signal.

TEEM predicts:

- additional quasi-normal modes associated with R-relaxation,
- delayed echo pulses following the main ringdown,
- slight deviations in damping times of standard GR modes.

These effects may be detectable by **LIGO/Virgo/KAGRA**, and will be especially testable with **LISA** and the **Einstein Telescope**.

7.5 Near-Horizon Anisotropies from R-Memory

Because the tension field Rretains long-memory of infalling matter, the near-horizon region may exhibit **persistent anisotropies**.

Observable consequences:

- small but measurable deviations in the polarization of emitted radiation,
- asymmetries in accretion flow structure,
- long-term perturbations in the gravitational field around the black hole.

These effects could be probed through:

- high-precision astrometry of stars orbiting Sgr A*,
- polarization mapping of accretion disks,
- long-baseline interferometry.

7.6 Summary of Distinguishing Predictions

TEEM-BH differs from GR in several measurable ways:

- **finite horizon thickness**,
- **larger and smoother shadow**,
- **non-thermal Hawking-like radiation**,
- **gravitational-wave echoes**,
- **near-horizon anisotropies**.

8. Conclusion

In this work, we have developed **TEEM-BH**, the black hole extension of the Triple Energy Exchange Model (TEEM3.0), and shown that black hole horizons, interiors, entropy, and information flow can be understood within a unified energetic framework. By treating spacetime not as geometric curvature but as an energetic configuration dominated by the spatial tension field R, TEEM provides a physically transparent and non-singular description of strong gravity.

We demonstrated that the event horizon corresponds to an **R-critical surface**, where the ratio v/Rgoverning time flow approaches zero. This energetic freeze-out reproduces the causal structure of GR while offering a deeper physical interpretation: the horizon is a **phase boundary** in the tension field, not a geometric singularity. Inside the horizon, gravitational collapse drives Rtoward its universal saturation value, forming a **finite-tension core** that replaces the classical singularity. This core is stable, finite, and information-bearing, consistent with the partial-reset mechanism that also governs cosmological bounces in TEEM3.0.

Black hole entropy arises naturally from **boundary tension modes** on the R-critical surface. Counting these microscopic degrees of freedom yields the area law and provides a physical interpretation of the Bekenstein–Hawking entropy coefficient as the density of tension modes, rather than a geometric Planck area.

Information is preserved through **R-reconfiguration**: infalling matter transfers its information into the tension field, which stores it in long-memory configurations and gradually releases it through correlated, non-thermal Hawking-like radiation as the black hole evaporates. This mechanism ensures full unitarity without requiring firewalls, exotic quantum gravity effects, or violations of semiclassical physics.

Finally, TEEM-BH yields several **testable predictions** that distinguish it from GR: a finite horizon thickness, modified shadow profiles, non-thermal components in Hawking-like radiation, gravitational-wave echoes from the R-critical layer, and near-horizon anisotropies arising from R-memory. These signatures fall within the reach of current and upcoming observational facilities, including the Event Horizon Telescope, LIGO/Virgo/KAGRA, LISA, and high-energy gamma-ray telescopes.

By grounding black hole physics in energetic rather than geometric principles, TEEM-BH provides a unified, non-singular, and observationally falsifiable framework for understanding strong gravity. It extends TEEM3.0 into the black hole regime and strengthens the case for spatial tension Ras the fundamental degree of freedom underlying gravitational, quantum, and cosmological phenomena.

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